VIN_RAW
VIN_FUSED
VIN_P
VIN_L
VBUS
+5V_BUCK
VSERVO
3V3_S
5V_S
J1
VIN_RAW
D1
SMBJ16A
VIN_FUSED
Q1
SUD50N06-15
VIN_FUSED
D2
12V
VIN_P
Q1_G
R1
100k
Q1_G
U1
TPS54608
VIN_P
BUCK_EN
COMP
RT/CLK
BOOT
SW
FB
C1
4.7uF
VIN_P
C2
4.7uF
VIN_P
C4
100nF
BOOT
SW
R2
2k
VIN_P
BUCK_EN
R3
68.7k
BUCK_EN
R4
43k
RT
R5
4.9k
COMP
C5
47nF
COMP_AC
C6
47pF
COMP
R6
4.64k
FB
+5V_BUCK
R7
2.2k
FB
D3
B340B
SW
L1
4.8uH
SW
+5V_BUCK
C7
47uF
+5V_BUCK
C8
47uF
+5V_BUCK
C9
47uF
+5V_BUCK
R8
10m
VSERVO
+5V_BUCK
U4
INA219A1
+5V_BUCK
VSERVO
SDA
SCL
C10
100nF
C11
470uF
VSERVO
C29
22uF
VSERVO
C30
22uF
VSERVO
Q2
2N7002
EN_KILL
BUCK_EN
Q3
2N7002
SERVO_EN
ESTOP_EN_KILL
R9
100k
VIN_L
EN_KILL
D4
SS14
EN_KILL
R10
100k
SERVO_EN
D5
SS34
VIN_L
VIN_P
C12
100uF
VIN_L
C13
10uF
VIN_L
U2
AMS3205
VIN_L
SW2
BST2
SW2
C14
100nF
SW2
BST2
L2
4.7uH
SW2
C15
22uF
C16
22uF
D6
SS14
VBUS
U3
AMS1117-3.3
C17
22uF
C18
10uF
FB1
600R
3V3_S
C19
10uF
3V3_S
F2
5V_S
400mA
5V_S
C20
10uF
5V_S
R11
100k
VIN_P
VIN_SENSE
R12
8k
VIN_SENSE
C21
100nF
VIN_SENSE
U5
ESP32-WROOM-32E
C22
10uF
C23
100nF
R13
10k
EN
C24
1uF
EN
SW1
EN
IO0
SW2
BOOT
R14
1k
LED_STAT
D7
STAT
LED_STAT
R15
1k
LED_3V3
D8
3V3
LED_3V3
R16
1k
VSERVO
LED_SERVO
D9
SERVO
LED_SERVO
J2
USB_C_16P
VBUS
CC1
CC2
USB_C_N
USB_C_P
SHIELD
GND
R17
5.1k
CC1
R18
5.1k
CC2
U6
USB_C_16P
USB_DP
USB_DN
R20
27R
R21
27R
U7
CH340R
USB_DP_S
USB_DN_S
ESP_RX
ESP_TX
CTS
DSR
DCD
DTR
RTS
C25
100nF
Q4
MMBT3904
RTS_EN
Q5
MMBT3904
DTR_IO0
R19
10k
DTR
Q4_B
R20
10k
RTS
Q5_B
U8
PCA9685
SCL
SDA
EXTCLK
OE
PCA_OE
LED0
LED1
LED2
LED3
LED4
LED5
LED6
LED7
LED8
LED9
LED10
LED11
LED12
LED13
LED14
LED15
A0
A1
A2
A3
A4
A5
VDD
VSS
PWM0
PWM1
PWM2
PWM3
PWM4
C26
100nF
C27
10uF
R21
10k
PCA_OE
R22
2.7k
SDA
R23
2.7k
SCL
U9
74HCT138
PWM4
PWM3
PWM2
PWM1
PWM0
SIG4
SIG3
SIG2
SIG1
SIG0
C28
100nF
S1
VSERVO
SIG0
R24
20
S1
S2
VSERVO
SIG1
R25
20
S2
S3
VSERVO
SIG2
R26
20
S3
S4
VSERVO
SIG3
R27
20
S4
S5
VSERVO
SIG4
R28
20
S5
DIO1_P
R30
1k
DIO1_P
3V3_S
R35
10k
3V3_S
C34
100nF
DIO1
DIO1_P
R31
1k
DIO2_P
DIO2
3V3_S
R36
10k
3V3_S
DIO2_P
C35
100nF
DIO2
DIO2_P
R32
1k
DIO3_P
DIO3
3V3_S
R37
10k
3V3_S
DIO3_P
C36
100nF
DIO3
DIO3_P
R33
1k
DIO4_P
DIO4
3V3_S
R38
10k
3V3_S
DIO4_P
C37
100nF
DIO4
DIO4_P
3V3_S
DIO5_P
R34
1k
DIO5_P
DIO5
3V3_S
R39
10k
3V3_S
DIO5_P
C38
100nF
DIO5
DIO5_P
3V3_S
DIO6_P
R40
1k
ADC1_P
ADC1
3V3_S
C39
100nF
ADC1
ADC1_P
3V3_S
ADC2_P
R41
1k
ADC2_P
ADC2
3V3_S
C40
100nF
ADC2
ADC2_P
3V3_S
ADC3_P
R42
1k
ADC3_P
ADC3
3V3_S
C41
100nF
ADC3
ADC3_P
3V3_S
ADC4_P
R43
1k
ADC4_P
ADC4
3V3_S
C42
100nF
ADC4
ADC4_P
3V3_S
SCL
SDA
3V3_S
SCL
SDA
5V_S
UART_TX
UART_RX
J21
ESTOP
H1
M3
H2
M3
H3
M3
H4
M3

Sheet: /
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